AUTOMATIC SEGMENTATION OF ICE SHELVES WITH DEEP LEARNING

Miguel Hoyo García, Elena Donini

Fondazione Bruno Kessler
University of Trento
Trento, Italy

Francesca Bovolo

Fondazione Bruno Kessler
Trento, Italy

ABSTRACT

Radar sounders (RSs) provide the information of the subsurface of the cryosphere through the use of electromagnetic (EM) signals by producing radargrams. Radargrams are used to detect and analyze relevant targets in the subsurface of icy regions. Up to now, studies of the subsurface structure of the cryosphere with radargrams has been conducted manually or applying semiautomatic techniques. However, these techniques present efficiency and adaptability disadvantages. To overcome these issues, we propose automatic analysis techniques for radargrams of icy regions based on deep learning (DL). Experimental analysis is conducted for the automatic segmentation of areas of interest in radargrams of ice shelves of coastal areas acquired by the radar sounder MCoRDS2.

Index Terms— radar sounder, subsurface sensing, deep learning, cryosphere, remote sensing

1. INTRODUCTION

RS refers to active nadir-looking remote sensors that probe the subsurface. These sensors transmit EM signals in the range bands of High Frequency (HF) and Very High Frequency (VHF). The signals are transmitted in the nadir direction and propagate through the subsurface. The transmitted signal interacts with the interfaces under the surface, and this information is collected by producing radargrams.

Radargrams of ice bodies have been mostly analyzed by visual interpretation. However, this has lately changed because of the eruption of machine learning techniques. Ilisei et al. [1] presented a method based on a Support Vector Machine (SVM) that automatically detects targets and segments inland radargrams. The method needs labeled training samples and several handcrafted input features, such as the Kullback–Leibler (KL) distance, the entropy, and a fitted model. Although the method obtains good results, the use of handcrafted features is a great inconvenience because they have to be adapted for every subsurface target.

Because of these limitations, the literature recently replaced SVMs with more advanced DL techniques such as convolutional neural networks (CNNs). The deep architecture of CNNs allows them to learn automatically features from the data and remove the limit of using hand-crafted input features. Generalization abilities improves as well as adaptation ones. These advantages make CNNs interesting for RS data processing. There is a large number of examples in the literature that address DL techniques. For instance, Ronneberger et al. [2] introduced the U-Net for segmentation of biomedical images. These images are quite similar to radargrams [2], which makes this network an interesting starting point for the interpretation of RS data. U-Net needs a labeled dataset for training, which is mostly unavailable for RS data. This issue can be avoided using unsupervised image segmentation techniques. In this field, Xia et al. [3] proposed the W-Net, a fully convolutional autoencoder (CAE).

In this paper, we present an approach for automatic segmentation of radargrams acquired in coastal areas by designing from scratch a CAE. The solution takes into account the specific characteristics of the targets in radargrams, such as the size or the location. The solution is able to automatically segment radargrams into five classes: air, inland ice, floating ice and crevasses, bedrock and noise without the need of handcrafted features and using a small amount of labeled data compared to other image segmentation approaches using CNNs.

The rest of this paper is organized as follows. Section 2 describes the proposed methodology. The results obtained on Antarctica coastal regions are described in Section 3. Finally, Section 4 presents conclusions and directions for future work.

2. PROPOSED METHODOLOGY

The subsurface information in radargrams allows studying the geological structure in the cryosphere, such as the floating area, which is crucial to understand the dynamics and evolution of the volume of an ice shelf caused by ice melting in the framework of climate change.

Cryosphere radargrams image sheets of ice laying over the continental land or ice floating on the ocean. The first encountered interface is generated by the ice surface which generates the strongest reflection in terms of power in the radargram. Below the surface, the ice layers generate bright lines as a function of the dielectric coefficient. In inland

© 2021 IEEE. Personal use of this material is permitted. Permission from IEEE must be obtained for all other uses, in any current or future media, including reprinting/republishing this material for advertising or promotional purposes, creating new collective works, for resale or redistribution to servers or lists, or reuse of any copyrighted component of this work in other works. Accepted manuscript. M. Hoyo García, E. Donini, F. Bovolo, IGARSS 2021, Brussels, Belgium. DOI: 10.1109/IGARSS47720.2021.9553610

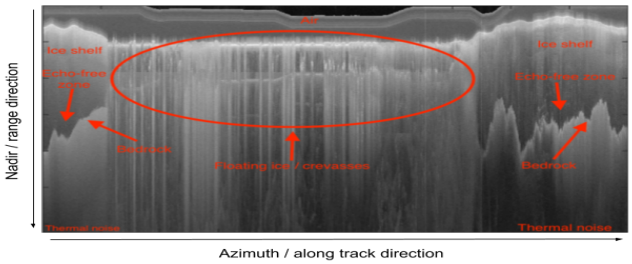


Fig. 1. Detail of a radargram of an ice shelf of a coastal area in Antarctica.

ice, the deepest reflection identifies the bedrock, which reflects all the remaining incident signal and produces a peak of the returned intensity. The noisy contributions in inland ice radargrams are identified in two areas. The first one is found over the bedrock, and it is known as the echo-free zone (EFZ) [4]. The EFZ has statistical properties similar to thermal noise. The second noisy area is located below the bedrock. As the bedrock reflects all the incidence signal, there is no information from the region below it. In radargrams of ice shelves of coastal areas, many targets can be identified, such as crevasses. Crevasses are vertical fractures in the floating caused by the movement of the ice shelf over the water for the ice melting. Crevasses are identified in the radargram as high scattered vertical lines produced by the penetration of the transmitted signal into them [5]. In floating ice radargrams, the features that cause the deepest reflection are the crevasses or the interface between floating ice and water when crevasses are not present. The noisy region in ice shelf radargrams is located below the interface between floating ice and water. Finally, common to all radargrams, a noisy area occurs over the surface because of free space or air. Fig. 1 shows an example of a radargram of a coastal area ice shelf from Antarctica.

Let the radargram $\mathbf{R}$ be a 2-dimensional matrix, with $[1, ..., n_T]$ traces along track or azimuth direction (columns) and $[1, ..., n_S]$ samples in the depth or range direction (rows):

$$\mathbf{R} = \{R(x, y) | x \in X = [1, ..., n_T], y \in Y = [1, ..., n_s]\} \quad (1)$$

The proposed model is a CAE that automatically segments and classifies $\mathbf{R}$ into N classes $\Omega = \{\omega_{1,}, \omega_2, ..., \omega_N\}$ using DL. Where $\omega_1 = \omega_a$ corresponds to the region above the surface, and $\omega_s = \{\omega_2, ..., \omega_N\}$ are the classes in the subsurface.

Fig. 2 shows the high-level block diagram of the proposed method. It is divided into two main steps: unsupervised pretraining and supervised training refinement. The aim of the first step (upper part of the block diagram) is to pretrain the weights of the CAE without any labeled dataset. The weights of the network are updated by applying the gradients calculated from two losses: the reconstruction loss (L_r), and

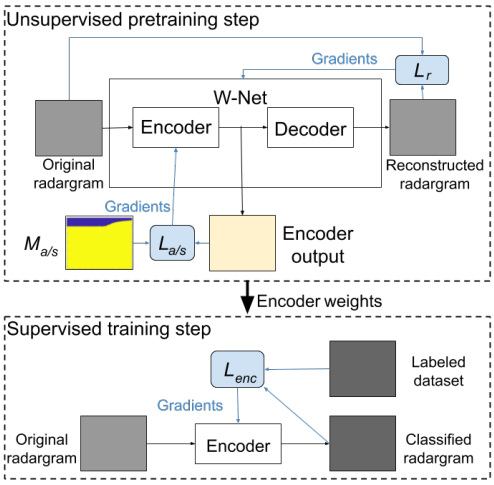


Fig. 2. Block scheme of the proposed method for the CAE.

the above/subsurface loss ($L_{a/s}$). In the supervised training refinement (bottom part of the block diagram), we tune the training of the pretrained CAE encoder using a small labeled dataset (generated by photointerpretation) compared with other approaches of image segmentation with CNNs. To optimize the process, we introduce weight maps to compensate the unbalanced prior probability of classes. As a result, the encoder automatically segments radargrams of ice bodies of coastal areas.

The proposed DL architecture is based on the W-Net [3]. This network is a CAE composed by two concatenated U-Nets [2]. The first U-Net acts as encoder and outputs a segmented radargram. The second U-Net acts as decoder and outputs the reconstructed radargram.

In the unsupervised pretraining step of the W-Net, the CAE reconstructs the input radargram R by minimizing the error between the radargram reconstructed by the W-Net ($Wnet(R)$) and $\mathbf{R}$:

$$L_r = ||R - Wnet(R)||^2 \quad (2)$$

L_r calculates the gradients applied to update all the weights of the W-Net.

$L_{a/s}$ is an additional loss used to update the encoder weights in the unsupervised pretraining step. To calculate $L_{a/s}$, a preliminary labeled data, the above/subsurface mask $M_{a/s}$ is used. The mask divides the original radargram in two classes: the area above the surface (ω_a), and the subsurface (ω_s). The surface of the original radargram is used as a rough boundary between the two classes. Assuming that the ice surface S has a maximum intensity in the preprocessed radargram in each range line, the surface corresponds to the row $[1, 2, ..., n_S]$ where the maximum value of each column $[1, 2, ..., n_T]$ appears: $S(x) = \underset{y}{max}\{R(x, y)\}$ [1]. Once the surface coordinates are calculated, $M_{a/s}$ is constructed as follows:

$$M_{a/s}(x,y) = \begin{cases} 0, & if\ 1 < y < R(x,y) = S(x) \\ 1, & elsewhere \end{cases} \quad (3)$$

$L_{a/s}$ calculates the error between the binary mask $M_{a/s}$ and the radargram segmented by the encoder $U_{enc}(R)$, after the SoftMax layer of the W-Net. $U_{enc}(R)$ contains more than two classes: above the surface (ω_a) and subsurface classes $\{\omega_2, ..., \omega_N\}$. So, a binary segmented radargram ($U_{enc}(R)_s$) that contains the probabilities of the pixels belonging to a subsurface class (ω_s) is obtained from $U_{enc}(R)$. The probability of ω_s corresponds to the sum of the subsurface classes $\{\omega_2, ..., \omega_N\}$ probabilities in $U_{enc}(R)$. The error between both $U_{enc}(R)_s$ and $M_{a/s}$, which is $L_{a/s}$, is calculated by the binary cross entropy loss. Let assume the subscript i as each possible combination of (x, y):

$$L_{a/s} = -\frac{1}{N_p} \sum_{i=1}^{N_p} M_{a/s_i} \cdot log(U_{enc}(R)_{s_i}) + \\ + (1 - M_{a/s_i}) \cdot log(1 - U_{enc}(R)_{s_i}) \quad (4)$$

where $N_p = n_S n_T$ is the number of pixels in $\mathbf{R}$. This loss assumes an improvement in the pretrained weights of the encoder for the final training step. The pretraining process stops when L_r and $L_{a/s}$ are minimized.

The supervised refinement of the pretrained U-Net encoder uses a minimal training set T, where the values of the radargram R are associated with labels LD. So $T = \{R(x,y), LD(x,y)\}$, $R(x,y)$ is the value of the radargram in each pixel associated with the label $LD(x,y)$, and $LD(x,y)$ may assume values in $[1, 2, 3, ..., N]$. The radargram segmented by the U-Net ($U_{enc}(R)$) is compared with $LD(x,y)$ to obtain the loss of the encoder (L_{enc}).

The process is optimized by the weight map that compensates the extremely unbalanced prior probability of the classes in the considered problem. For example, the bedrock class is up to twenty times less frequent than the noise or the floating ice ones. Thus the network hardly learns to differentiate these less frequent classes because their contribution in the overall value of the loss is low. The weight of a class (w_{ω_n}) depends on the prior probability of the class that the pixel belongs to in T:

$$w_{\omega_n} = \frac{N_p}{N_{\omega_n} N} \quad (5)$$

where N_{ω_n} is the total of pixels of ω_n in T. Finally, w_{ω_n} contributes to calculate L_{enc}, which is based on the sparse categorical cross entropy loss:

$$L_{enc} = -\frac{1}{N_p} \sum_{i=1}^{N_p} \sum_{n=1}^{N} y_{LD_i,\omega_n} \cdot log(U_{enc}(R)_{i_{\omega_n}}) w_{\omega_n} \quad (6)$$

Table 1. Parameters and characteristics of MCoRDS2 [6].

Parameter	Value
Central frequency (f_c)	193.9 MHz
Bandwidth (BW)	30 MHz
Transmitted power (P_{tr})	1050 W
Aircraft altitude (h)	500 m
Range resolution in Ice (R_r)	4.3m
Along-track resolution (R_a)	25 m

Table 2. Parameters of the dataset.

Parameter	Patches
Unsupervised training	1173+750
Supervised training	1173
Validation (supervised & unsupervised step)	234
Test	410

where y_{LD_i,ω_n} is a binary indicator $[0, 1]$ that its value is 1 when LD_i is the same class as ω_n. $U_{enc}(R)_{i_{\omega_n}}$ refers to the probability of ω_n in the pixel i. This loss is used to update the weights of the encoder.

3. EXPERIMENTAL RESULTS

The dataset includes eight radargrams from four acquisition campaigns in coastal areas of the Antarctica provided by the Center of Remote Sensing of Ice Sheets (CReSIS): 20161109_03, 20181015_01, 20181115_01 and 20181012_01. The radar is the MCoRDS2 on a Douglas DC-8 aircraft [6]. The parameters of the radar are shown in Table 1.

The data are azimuth focused via synthetic aperture radar (SAR) processing. We corrected the fluctuations caused by the movements of the aircraft and the intensity information of each radargram was expressed in decibels and normalized. Finally, each radargram was divided into vertical patches of 1536x64 pixels to adapt the data to the W-Net requirements. The patch size increases the variability in terms of the considered classes: air, layers of the ice sheet, floating ice, bedrock, and EFZ and thermal noise ($\Omega = \{\omega_{air}, \omega_{layers}, \omega_{bedrock}, \omega_{floating\,ice}, \omega_{noise}\}$). For the unsupervised pretraining, 4 of the radargrams are employed for the training set (1923 patches) and 2 for the validation set (234 patches). For the supervised training refinement, 3 radargrams of the unsupervised training set (1173 patches) are labeled alongside the validation set and the other two remaining radargrams, which are employed for test. Table 2 shows the dataset parameters.

Both the W-Net unsupervised pretraining and the supervised W-Net encoder training last 200 epochs with a learning rate of 0.001 and the Adam optimizer.

Table 3. Error matrix of U-Net on the test samples.

		Reference pixels					
	Class	Air	Noise	Bedrock	Layers	Floating ice	AUA
Pred. value	Air	9606241	8162	0	7571	9749	99.74%
	Noise	0	20833519	217812	306013	990977	93.22%
	Bedrock	0	46022	313098	180	2200	86.61%
	Layers	2039	559693	1069	3712530	40927	86.01%
	Floating ice	13903	114897	5989	47016	3511276	95.08%
	APA	99.83%	96.62%	58.20 %	91.14%	77.08%	OA = 94.12%

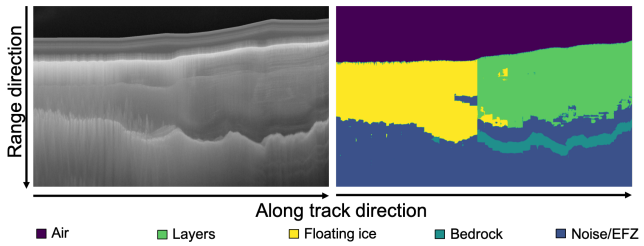


Fig. 3. Original radargram on the left and its segmentation map by the U-Net on the right.

The overall accuracy (OA) in the test dataset on a pixel basis is over the 94% (see Table 3). This proves that the method is able to effectively identify and segment the geological structure of radargrams of coastal ice bodies. The classes show different behaviors in the average producer accuracy (APA), from 58% of the bedrock to a 99% of the air class. Regarding the average user accuracy (AUA), the classes reach a better performance, from an 86% of the layers to a 99% of the air class. Since the majority of the test samples were taken from a different campaign than the samples used for training, this also shows the high robustness of the approach. Fig. 3 shows a portion of a segmented radargram from the test set. The visual analysis shows a good segmentation quality, in particular how the network properly delimitates the different classes of the radargram. It is important to notice how the network differentiates the floating ice (95.08% AUA) and the layers of the continental part of the ice sheet (86.01% AUA), since the visual characteristics of both regions are very similar and differentiating them is complex. Also, the identifiaction of the floating ice could help the measurements of the floating ice thickness.

4. DISCUSSION AND CONCUSSIONS

This paper proposes a technique to automatically identify the classes of radargrams acquired in the cryosphere. Including the floating ice, which is identified by the network and could be employed to estimate its thickness in different periods. This is key to estimate the evolution of the volume of an ice shelf caused by ice melting in the framework of climate change since the floating ice is the most sensitive area. A CAE is trained from scratch and uses a small amount of labeled data for training with respect to other image segmentation approaches using CNNs. The CAE reaches a high performance in the segmentation of radargrams by their geological structure into five classes.

As future work, we plan to compare this approach with other pretrained DL architectures using transfer learning techniques to adapt them to this domain. In addition, we aim to demonstrate its effectiveness on a larger number of classes in radargrams of both icy and arid areas.

References

[1] A. M. Ilisei and L. Bruzzone, "A system for the automatic classification of ice sheet subsurface targets in radar sounder data," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 53, no. 6, pp. 3260–3277, 2015.

[2] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *International Conference on Medical image computing and computer-assisted intervention*, Springer, 2015, pp. 234–241.

[3] X. Xia and B. Kulis, "W-net: A deep model for fully unsupervised image segmentation," *arXiv preprint arXiv: 1711.08506*, 2017.

[4] G. d. Q. Robin and D. Millar, "Flow of ice sheets in the vicinity of subglacial peaks," *Annals of Glaciology*, vol. 3, pp. 290–294, 1982.

[5] L. Lindzey, *A brief introduction to ice-penetrating radar*, 2015. [Online]. Available: `https://lindzey.github.io/`.

[6] L. Shi, C. T. Allen, J. R. Ledford, F. Rodriguez-Morales, W. A. Blake, B. G. Panzer, S. C. Prokopiack, C. J. Leuschen, and S. Gogineni, "Multichannel coherent radar depth sounder for nasa operation ice bridge," in *2010 IEEE International Geoscience and Remote Sensing Symposium*, IEEE, 2010, pp. 1729–1732.